

Robert D. McLaughlin
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10 CFR 50.73

LR-N24-0027

March 19, 2024

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-001

Hope Creek Generating Station
Renewed Facility Operating License No. NPF-57
Docket No. 50-354

Subject: Licensee Event Report 2023-003-01, Inadvertent Main
Turbine Control Valve Closure Caused Reactor Scram

PSEG Nuclear LLC submits the attached licensee event report supplement per 10
CFR 50.73(a)(2)(iv)(A).

There are no regulatory commitments contained in this submittal.

Please contact Ms. Bernadette Humphrey at (856) 339 - 1308 if you have questions.

Sincerely,

A handwritten signature in black ink that reads "Robert D. McLaughlin". The signature is fluid and cursive, with the first letters of the first and last names being capitalized and prominent.

Robert D. McLaughlin
Plant Manager, Hope Creek Generating Station

Attachment: Licensee Event Report

cc: USNRC Regional Administrator – Region 1
USNRC NRR Project Manager – Hope Creek
USNRC Senior Resident Inspector – Hope Creek
NJ Department of Environmental Protection, Bureau of Nuclear Engineering

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Hope Creek Generating Station	☒ 050	2. Docket Number 00354	3. Page 1 OF 3
	☐ 052		

4. Title
Inadvertent Main Turbine Control Valve Closure Caused Reactor Scram

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
12	14	2023	2023	- 003 -	01	03	19	2024	Facility Name	☐ 052	Docket Number

9. Operating Mode 1	10. Power Level 88
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).**12. Licensee Contact for this LER**

Licensee Contact Bernadette Humphrey	Phone Number (Include area code) (856) 339 - 1308
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On December 14, 2023, at 1939 EST, while attempting to manually close a failed open #4 turbine control valve (TCV), the #4 TCV closed rapidly. This caused a scram on average power range monitor (APRM) high neutron flux due to a rise in reactor pressure.

Hope Creek Generating Station (HCGS) repaired the #4 TCV by replacing the servo valve. In addition, the fast-acting solenoid (FAS) valve and the shutoff valve were replaced. HCGS synchronized the Main Generator to the grid on December 17, 2023, at 1537 EST.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME

Hope Creek Generating Station



050



052

2. DOCKET NUMBER

00354

3. LER NUMBER**YEAR**

2023

**SEQUENTIAL
NUMBER**

003

**REV
NO.**

01

NARRATIVE**ABSTRACT (continued)**

A Root Cause Evaluation (RCE) was performed. The root cause of this event was inadequate adherence to applicable administrative tools and processes. Corrective actions include procedure revisions and reinforcement of expectations regarding procedure adherence.

HCGS reported this per 10 CFR 50.72(b)(2)(iv)(B) as event notification 56889 and now reports this event per 10 CFR 50.73(a)(2)(iv)(A).

PLANT AND SYSTEM IDENTIFICATION

General Electric - Boiling Water Reactor (BWR/4)
Turbine Control Valve/ Electro-Hydraulic Control System (EHC) - TG

IDENTIFICATION OF OCCURRENCE

Event Date: December 14, 2023
Discovery Date: December 14, 2023

CONDITIONS PRIOR TO OCCURRENCE

OPCON 1, POWER OPERATION. Reactor power was at 88 percent.

No other structures, systems or components that could have contributed to the event were inoperable during the event.

DESCRIPTION OF OCCURRENCE

On December 11, 2023, an EHC leak was discovered on the #4 turbine control valve (TCV). Reactor power was reduced to 90 percent, the #4 TCV was closed, and EHC was isolated to stop the leak while the plant was at 90 percent power. HCGS further lowered power to 70% to facilitate conditions for repairs.

On December 14, 2023, HCGS was at 88 percent power after #4 TCV FAS oil supply tubing repairs. During restoration, foreign material that entered and damaged the servo o-rings during the leak repair caused the #4 TCV to go full open despite having a closed demand signal. While taking action to manually close the #4 TCV, the valve unexpectedly closed in approximately one second causing an automatic actuation of the reactor protection system (RPS) due to high average power range monitor (APRM) high neutron flux.

All control rods fully inserted into the core. All safety systems responded as designed and expected. There were no emergency core cooling system actuations during the event.



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1. FACILITY NAME	☒ 050 ☐ 052	2. DOCKET NUMBER	3. LER NUMBER		
Hope Creek Generating Station		00354	YEAR 2023	SEQUENTIAL NUMBER - 003	REV NO. - 01

NARRATIVE

CAUSE OF EVENT

Rapid closure of the #4 TCV caused a scram on APRM high neutron flux.

The direct cause of the full open #4 TCV was foreign material exclusion (FME) practices were not implemented correctly by the work groups. Decision makers believed manually operating the FAS could be done in a controlled manner; however, the subsequent fast closure of the #4 TCV resulted in the reactor scram.

The human performance related cause of this event was inadequate adherence to applicable administrative tools and processes. Vendor information was acted upon without employing a formal decision-making process.

SAFETY CONSEQUENCES AND IMPLICATIONS

There were no safety consequences because of this event. There was no radiological release. All safety equipment functioned as designed.

PREVIOUS EVENTS

There have been no previous events at HCGS in the last three years where the #4 TCV caused a reactor scram.

CORRECTIVE ACTIONS

1. Hope Creek Generating Station (HCGS) repaired the #4 TCV by replacing the servo valve, the fast-acting solenoid valve, and shutoff valve. HCGS synchronized the Main Generator to the grid on December 17, 2023, at 1537 EST.
2. Revise applicable EHC procedures which involve manual FAS valve manipulations while on-line.
3. Conduct training to review roles and responsibilities of the station duty team during the response to emergent issues.

Additional actions are being tracked in the licensee's Corrective Action Program.

COMMITMENTS

There are no regulatory commitments in this LER.